

Healthcare Analytics with SQL

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Project Objective

- Design a normalized healthcare database using MySQL.
- Establish relationships between healthcare entities using Primary and Foreign Keys.
- Import healthcare data into MySQL tables.
- Perform SQL analysis to extract meaningful business insights.
- Support hospital decision-making using data.



Database Design

Created five relational tables:

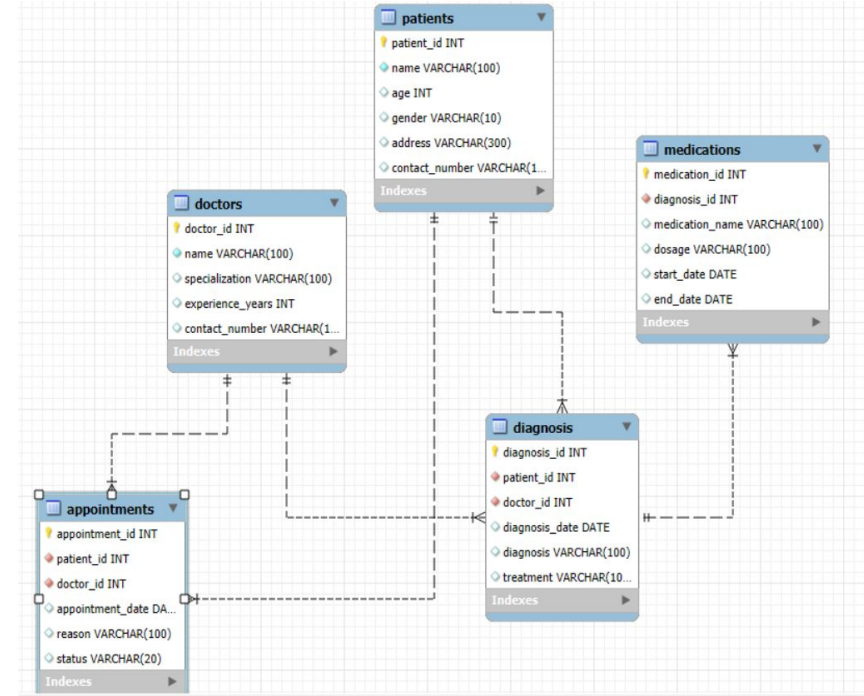
- Patients
- Doctors
- Appointments
- Diagnosis
- Medications



ER Diagram

Key Relationships

- One patient can have multiple appointments.
- One doctor can treat multiple patients.
- One appointment can have one diagnosis.
- One diagnosis can have multiple medications.



Task

Retrieve patient details along with doctor name, appointment date, and appointment status.

Insight

- Only patients with valid appointments were displayed.
- Successfully combined data from multiple related tables.

Result Grid					Filter Rows:	Export:	Wri
	Patient_Name	Doctor_Name	appointment_date	status			
▶	Patient_1	Doctor_214	2023-11-02	Cancelled			
	Patient_1	Doctor_205	2024-03-31	Cancelled			
	Patient_2	Doctor_57	2023-02-14	Scheduled			
	Patient_3	Doctor_94	2024-06-26	Cancelled			
	Patient_3	Doctor_39	2024-01-07	Completed			
	Patient_4	Doctor_77	2022-02-26	Cancelled			
	Patient_5	Doctor_206	2023-09-16	Completed			
	Patient_6	Doctor_17	2022-10-06	Scheduled			
	Patient_6	Doctor_122	2022-01-26	Cancelled			
	Patient_7	Doctor_69	2024-03-13	Completed			
	Patient_8	Doctor_132	2024-03-22	Completed			
	Patient_8	Doctor_85	2023-01-23	Cancelled			
	Patient_8	Doctor_191	2024-10-10	Cancelled			
	Patient_9	Doctor_106	2024-06-14	Completed			

Task

Find patients who have never had an appointment.

Insight

- Identified patients who have never booked an appointment.
- Displayed patient name, contact number, and address.

name	contact_number	address
Patient_11	98765430011	Address_11
Patient_22	98765430022	Address_22
Patient_35	98765430035	Address_35
Patient_36	98765430036	Address_36
Patient_45	98765430045	Address_45
Patient_47	98765430047	Address_47
Patient_55	98765430055	Address_55
Patient_58	98765430058	Address_58
Patient_76	98765430076	Address_76
Patient_83	98765430083	Address_83
Patient_87	98765430087	Address_87
Patient_94	98765430094	Address_94
Patient_104	98765430104	Address_104
Patient_109	98765430109	Address_109

Task

Count the number of diagnoses handled by each doctor.

Insight

- Displayed every doctor's specialization.
- Calculated the total diagnoses handled by each doctor.
- Doctors without diagnoses would also be included.

name	specialization	total_diagnoses
Doctor_1	Orthopedics	48
Doctor_2	Pediatrics	48
Doctor_3	Cardiology	59
Doctor_4	Pediatrics	60
Doctor_5	Cardiology	58
Doctor_6	Orthopedics	57
Doctor_7	Cardiology	55
Doctor_8	General Medicine	46
Doctor_9	Pediatrics	47
Doctor_10	General Medicine	53
Doctor_11	Orthopedics	53
Doctor_12	Pediatrics	46
Doctor_13	Orthopedics	39
Doctor_14	General Medicine	43

Task

Compare appointment records with diagnosis records.

Insight

- Compared appointments and diagnoses.
- Identified records without matching diagnoses.

appointment_id	patient_name	doctor_name	diagnosis
1	Patient_4219	Doctor_5	NULL
2	Patient_2182	Doctor_202	NULL
3	Patient_1643	Doctor_202	NULL
4	Patient_3889	Doctor_179	NULL
5	Patient_841	Doctor_57	NULL
6	Patient_2655	Doctor_186	NULL
7	Patient_4119	Doctor_92	NULL
8	Patient_566	Doctor_292	NULL
9	Patient_965	Doctor_227	NULL
10	Patient_2283	Doctor_112	NULL
11	Patient_996	Doctor_148	NULL
12	Patient_1188	Doctor_269	NULL
13	Patient_4922	Doctor_203	NULL
14	Patient_4312	Doctor_56	NULL

Task

Rank patients for each doctor based on appointment count..

Insight

- Ranked patients separately for each doctor.
- Patients with equal appointment counts received the same rank.
- Only one appointment made by all the patients.

doctor_id	patient_id	total_appointments	patient_rank
1	3365	1	1
1	1551	1	1
1	2949	1	1
1	744	1	1
1	4830	1	1
1	624	1	1
1	1936	1	1
1	3433	1	1
1	1577	1	1
1	3512	1	1
1	96	1	1
1	1362	1	1
1	4834	1	1
1	874	1	1

Task

Categorize patients based on age.

Age Groups

- 18 to 30
- 31 to 50
- 51+

Insight

- 51+ years: **2684 patients**
- 31 to 50 years: **1416 patients**
- 18 to 30 years: **900 patients**

Observation:

The majority of patients belong to the 51+ age group.

Age_Group	Total_Patients
51+	2684
31-50	1416
18-30	900

Task

Retrieve patients whose contact number ends with **1234**.

Insight

- Successfully identified matching patients.
- Demonstrated the use of pattern matching and string functions.

Patient_Name	contact_number
PATIENT_1234	98765431234

Task

Find patients prescribed **Insulin**.

Insight

- Identified patients receiving Insulin treatment.
- Eliminated duplicate records using DISTINCT.

name
Patient_1274
Patient_4921
Patient_1545
Patient_1440
Patient_4206
Patient_1600
Patient_3815
Patient_4331
Patient_157
Patient_3365
Patient_466
Patient_1595
Patient_3313
Patient_3087
Patients 8 ×
put

Task

Calculate the average medication duration.

Insight

- Calculated average treatment duration for each diagnosis.
- Demonstrated the use of SQL date functions.
- Negative values detected.

diagnosis_id	Average_Duratio
1	-245.3333
4	372.0000
6	652.0000
8	439.0000
9	-77.0000
11	-34.0000
12	-134.5000
13	-216.0000
14	-199.0000
15	-223.0000
17	-14.0000
18	10.0000
20	-294.0000
21	-343.0000



Task

Identify the doctor who attended the highest number of unique patients.

Insight

- **Doctor_37** (General Medicine) attended the highest number of unique patients.
- **Unique Patients: 51**

	name	specialization	Unique_Patients
	Doctor_37	General Medicine	51

Additional Tasks

Most Common Diagnosis: *Hypertension* (184 cases)

Most Prescribed Medication: *Paracetamol* (256 prescriptions)

Busiest Specialization: *General Medicine* (412 appointments)

Appointment Status: *Completed: 730 | Scheduled: 210 | Cancelled: 60*



Key Insights

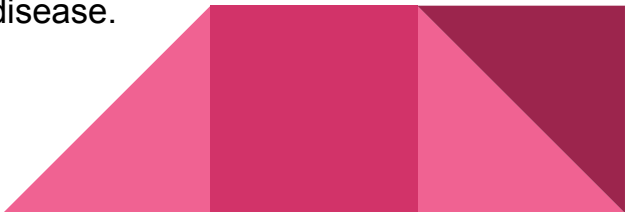
Patient Demographics

- The **51+ age group** had the highest number of patients (**2,684**), followed by the **31-50** age group (**1,416**) and the **18-30** age group (**900**).

Doctor Performance

- **Doctor_37 (General Medicine)** treated the highest number of unique patients (**51**), making them one of the busiest doctors in the dataset.

Most Common Diagnosis

- The analysis identified **Hypertension** as the most frequently diagnosed disease.
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Most Prescribed Medication

- **Paracetamol** was the most frequently prescribed medication.

Busiest Medical Specialization

- **General Medicine** recorded the highest number of appointments among all departments.

Appointment Status Analysis

- Appointment status analysis showed the distribution of **Completed, Scheduled, and Cancelled** appointments.
- Cancelled appointments are very less compared to completed appointments. It directly shows the efficient scheduling Method of the healthcare institution.

Patient Visit Pattern

- The ranking analysis showed that **every patient had only one recorded appointment**, resulting in all patients receiving **Rank 1**.



Recommendation

- Strengthen follow-up programs for patients aged **51 years and above**, as they represent the largest patient group.
 - Allocate additional resources to the busiest specialization to reduce patient waiting times and improve service quality.
 - Ensure sufficient stock of the **most prescribed medication** to avoid shortages.
 - Monitor the **most frequently diagnosed disease** and conduct preventive health awareness programs where appropriate.
 - Review cancelled appointments regularly and implement reminder systems to improve appointment attendance.
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THANK YOU

